

Advanced Database Management System Format Lab Project Guidelines

Course: Advanced Database Management System Lab

Technology Focus: MongoDB and Modern NoSQL Databases

Group Size: 3 Members

Submission Format: PDF Report + Project Files

The goal of this project is to help students apply advanced database concepts by designing and developing a complete NoSQL database application using **MongoDB** as the primary backend. Students are expected to demonstrate practical skills in document database design, CRUD operations, aggregation, indexing, optimization, transactions, concurrency, security, and scalability.

Project Requirements

1. Backend: MongoDB

- Design and implement a MongoDB database for a real-world application.
- Create a minimum of **4 related collections** with appropriate document structures.
- Demonstrate the use of:
 - CRUD operations
 - Embedded documents and/or references
 - Aggregation pipelines
 - Indexing strategies
 - Query optimization techniques
 - Transactions for data consistency
- Apply appropriate validation rules and schema design principles.

2. Frontend / Application Interface

- Develop a user interface using any suitable technology, such as:
 - HTML/CSS with PHP
 - JavaScript (Node.js/React)
- The application should allow users to:
 - Insert, update, and delete records
 - Search and retrieve data
 - View filtered and sorted results
 - Generate analytical reports or summaries

3. Advanced MongoDB Features

Your project must incorporate the following advanced features:

- Aggregation Pipeline for reporting and analytics
- Indexing and performance optimization
- Multi-document transactions

- Concurrency handling mechanisms
- Views or materialized views
- Database security and access control
- Sharding or scalability concepts
- Integration with MongoDB Atlas

4. Optional Comparative Component (Bonus)

- You may include a brief comparison or prototype using another NoSQL database such as:
 - Amazon DynamoDB
 - Google Firestore
 - Azure Cosmos DB

Project Report (Minimum 12 Pages, Maximum 18 Pages)

Submit a professionally formatted PDF report containing the following sections:

a. Title Page

Provide basic project details including title, group members, and course information.

b. Table of Contents

List all sections of the report with page numbers for easy navigation.

c. Introduction

Briefly explain the problem, background, and why the system is needed.

d. Objectives

State the main goals and what your project aims to achieve.

e. Requirements Analysis

Describe system requirements, including what the system should do (functional) and performance/security needs (non-functional).

f. MongoDB Implementation

Present key MongoDB operations including collection creation, CRUD queries, aggregation pipelines, and transactions with proper explanation.

g. Performance Optimization

Explain how indexing and query optimization techniques were used to improve performance.

h. Security and Scalability

Describe how the system ensures data security, access control, and handles growth using scalability techniques.

i. Application Interface

Show and explain the user interface, including main features and system workflows.

j. GitHub Repository

Each group member must upload the complete project to their own GitHub repository and provide the repository link in the report. All repositories should contain the same final project with proper structure and documentation. Projects without a properly maintained GitHub repository may lose marks.

k. Challenges and Lessons Learned

Discuss difficulties faced during development and what you learned from solving them.

l. Conclusion and Future Enhancements

Summarize the project outcomes and suggest possible improvements or future work.

Final Submission Package

Submit a compressed .zip folder containing:

- Final Project Report (PDF)
- Source Code Files (Frontend + Backend)
- MongoDB Scripts / Export Files
- README File (steps to run the project)
- GitHub Repository Link (included in report)

Project Demonstration

During the final evaluation, each group must:

- Demonstrate the working application
- Showcase MongoDB features used (CRUD, aggregation, indexing, etc.)
- Execute sample operations live
- Answer questions related to implementation and design

Evaluation Criteria

Criteria	Marks
Database Design and Schema Modeling	10
MongoDB Implementation (CRUD & Queries)	15
Aggregation Pipeline and Advanced Features	10
Indexing and Query Optimization	10
Transactions, Concurrency Control, Security, and Scalability	15
Frontend Integration and Functionality	15
Report Quality and Documentation	10
Project Demonstration and Viva	10
GitHub Repository and Code Management	5
Total	100

Project Ideas:

Students may choose their own project domain, provided it demonstrates meaningful application of MongoDB and advanced database concepts. The following ideas are provided for inspiration:

1. Smart Healthcare Record Management System
2. E-Commerce Product Catalog and Recommendation System
3. Online Food Delivery and Order Tracking System
4. Library Management and Book Recommendation Platform
5. Real Estate Property Listing and Analytics System
6. Student Performance Analytics and Academic Dashboard
7. Inventory and Supply Chain Management System
8. Social Media Content and User Interaction Analysis Platform
9. Event Management and Ticket Booking System
10. Ride-Sharing and Route Optimization System
11. Hotel Reservation and Customer Feedback Analysis System
12. Online Learning Management and Progress Tracking System
13. Fraud Detection and Transaction Monitoring System
14. Multi-Vendor Marketplace Management System

Note: Students are encouraged to select projects that can effectively incorporate concepts such as aggregation pipelines, indexing, query optimization, transactions, concurrency control, sharding, security, and cloud-based NoSQL services.